

Weekly Study notes: Week 11

Presentation Tips:

- We have spoken about not being too abstract in your presentation and tie your presentation back to Sovereign Bank. In some cases, depending on the topic, you don't need to twist your presentation too much. Think about adding more examples that can be relevant to Sovereign Bank. In general, think always about what is in for the customer.
- The Customer Outcomes component is key. Go through the weekly challenges, key considerations and desired outcomes and see how you can address those in your presentation. Continuously link back to the challenges and desired outcomes, not just at the beginning of your presentation summarising them.
- How to handle questions? Be succinct, address the main points.

Testing Leading Practices (including info from previous cohorts):

- Testing is commonly an afterthought, and every project requires a different testing strategy.
- * As we discussed testing can summed up as "prevention is better than cure". Poor testing will result in bugs/defects which delivers a lower level of service with regard to application development/service.
- Testing should follow several paths. Test best case, test the way something should work, as well as test error handling and failure modes. In other word test for the representative, marginal and extreme situations.
- Testing needs to consider more than WHAT gets tested and WHEN. It's important to also consider WHO does the testing (may be different in different stages). A dedicated team helps prioritize testing/governance. Could be a third-party.
- Trying to get someone to test, when it's not their day job, is tricky. Need training and good documentation. Make sure they understand the platform.
- Training the testers can easily become "do it this way" advice to that team... which runs counter to exploratory testing and experience feedback.
- If the first budget cut is testing, the first question asked after release almost always is "who tested this?".
- Developers aren't necessarily good testers. They may unconsciously test their code in a way that works (the happy path).
- * We also discussed that automated testing may take time/and money to develop but should pay-off in the long run.
- Fit development and testing in the sprint (unit & peer). May need to do easier stories early

in the sprint so it doesn't build up toward the end. Keep end sprint light to get to backlog for testing feedback. Developer and tester need to be in lockstep from beginning of story.

- Most orgs' QA/UAT team will be well-versed in the behaviour of their existing tool and will greatly judge SN against that tool and process – effective OCM helps.

* We also discussed that setting up expectations with stakeholders is important as it can often mean that defined behaviour is not seen as a defect because it does not match legacy tools. Having clear definitions between defects and enhancements is key to enable expectations in stakeholders.

- Process Owners are great for testing, because they 'know' the process, and can determine if the test/test failure is due to the tool or due to the process. Having said that process owners need to be 'on-board' with the new process transformation process, particularly in relation to the situation of Sovereign Bank.

Performance Optimisation (including info from previous cohorts):

- Discuss optimisation when the developers are onboarded.
- Retention can be challenging in the public sector.
- Table Rotation can be problematic if you are in a regulated industry that requires record retention for a certain period of time.
- Requested items are important for retention (wanting to know who's requested and approved what)
- Don't want until there is a performance issue to discuss archiving, etc. However, it can be hard to plan for future table growth and when/how it will impact performance.
- Optimization is much easier if done early. It is much harder when the table size has already grown and inefficient code is already in production.
- Indexing for performance is our best friend. But also remember that indexes that are poorly applied can actually degrade performance.

Go-Live Preparation:

- Go-live planning should be an earlier stage with OCM.
- * I think we can agree that platform change, process change and Go-Live are as much a 'hearts and minds' challenge as a technical challenge (and arguably more so). Communication and involvement (keep your friend close...) is key.
- Involving managers early and often helps ease concerns of disruptions of go live.
 - Sometimes managers/process owners feel threatened if the original process was their design.
 - Understand concerns of end users (how will this impact my job? Will it be as painful as the last change?). People can be informed and consulted even if they are not the decision makers.
 - Value of the platform – this thing we are rolling out, will help you with roll out.

- Enable process owners and super users on the platform - they will own communications around process changes after implementation. Socialize the change.
- Include screenshots and videos in communications/resources.
- Do a demo for the process owner/team manager so they can see the improved efficiency. Long-term improvement is (hopefully) worth short-term shift & change.
- Rolling something back and correcting it may not be the right answer as some issues may occur only in production. In general, we do prefer to fix and go ahead with the rollout. If we have communicated the * Go Live and we roll back, we may have some reputational issues and people may lose confidence in this rollout. In general, we do not like rolling back.

Resources:

Table maintenance: <https://www.servicenow.com/community/now-platform-articles/demystifying-table-rotation-extension-and-table-cleaner/ta-p/2323658>

Story telling with data: <https://www.storytellingwithdata.com/>

Test Management presentation in Now

Create: https://nowlearning.servicenow.com/nowcreate/en/pages/assets?id=nc_asset&asset_id=bbcf6e5f87ee9d9024e0bb39dabb3502&nc_source=copy_asset_link

Weekly Study Notes (Week 10)

Presentation notes

- I have seen some great presentations over the course and it is really great to see the quality of the slides as well as the quality of the content. As I have stated before, presentations communicate information, persuade and invoke an emotional response. The metaphor I would use is going for a job interview. What you say is important but also how you present yourself is as important if not more important.
- After you develop your slides, go back and look at the slide titles and make sure they align to your agenda. Do the agenda and accompanying slides support the goal of your presentation?
- Don't forget a transition statement when you move to a new topic. Ideally it relates the topics together. It is ok to say "Next we will talk about...", but I'd like to challenge you to think more creatively (e.g. an linking anecdote or quote).

Design and Optimize Intelligent Workflows

- Process optimization leverages PA, Continuous Improvement and Machine Learning, all of which require a body of data to exist before they deliver value. It can take time to collect this data, and it can be helpful to use a maturity-based approach where these tools are leveraged more as more (and healthier) data exists.
- KPI composer and Process Optimization - great tools to be proactive, versus after implementation when leadership comes back looking for the data. Knowing ahead of time helps see what is most important to the org.
- Process Optimization requires a separate license... as always, check with the ServiceNow account manager for your customer.

Process Automation Designer

- Simplifies for process-oriented people, but relies heavily on the existence of appropriate building blocks (flows, subflows, actions)
- PAD allows good collaboration between personas. Designed to use the strengths of different people e.g. developers can focus on creating reusable process steps while process owners can focus on putting them together to design an end-to-end process.
- Can install PAD on PDI – it may take a while
 - Process Automation Designer for App Engine [com.glide.pad.license]
 - Playbooks for Customer Service Management [com.sn_csm_playbook]
- HR tables are not supported but you can use PAD on a case (use HR Service Delivery Playbook for lifecycle events)

Citizen Development

- There was some debate over whether Citizen Developers pose more of a net benefit or a net risk. On balance I would argue that Citizen Developers can be a benefit to the enterprise if their work is set within strong governance guardrails and governance. Citizen Developers need to go through a clear enablement process culminating in Systems Administration Certification. As stated the benefits of Citizen Developers are as follows:

- (i) Increase development capacity (taking on simple development tasks and requirements and freeing up the Sprint teams).
- (ii) Provide stakeholder 'buy-in'. If the stakeholders are involved, this will go a long way to reducing resistance and enable stakeholders to have a positive relationship to the platform and it's functioning.
- (iii) Provide a environment of ServiceNow knowledge and skills that can enable CDs to move to become full blown Developers.
- In my opinion, it might be worth involving Citizen Developers in 'safe' development activities such as:
 - (i) Creating testings for the Automated Test Framework application.
 - (ii) Creating required Reports and Dashboards
 - (iii) Creating Catalog Items and simple Flows.
- Center of Excellence and Innovation (CoEI) is a great starting point when discussing Citizen Development.
- Be careful on assumptions of skill levels of Developers vs Citizen Developers. The differentiation can also be one of job role rather than skill/experience.
- Teams this week covered the Enable, Equip and Control aspects of Citizen Development, but what about Support/Maintain?
 - People who want to extend/create new functionality are rarely interested in supporting/maintaining it
 - Who is responsible for the support/maintenance of capabilities created by CitDev?
 - Ties back to governance – what are Citizen Developers allowed to build in the first place (technical governance)? How does the Demand Management function fit in with Citizen Development?
 - Escalation from business to development works best when there is a Business Analysis function in between – this is the part that has understanding of both sides
 - Quote from Kevin: "Citizen developers are only concerned with their immediate environment, looking at the problem that they are trying to

solve so they can do their job, rather than seeing it in the context of the wider IT ecosystem.”

- How do you combine the people who want functionality but are not accountable for supporting it, with the people who have operational accountability but may not have capacity to build things through formal development processes?
 - Essential for there to be effective handover from CitDev to operational support teams. This is a good opportunity for checks against technical/quality expectations.
- For Sovereign Bank, Citizen Development is phase 2 or 3. For now, just introduction to the program and possibilities. BUT setting it up for success by building out app templates and flow templates would help accelerate the process once they are ready.
 - Need to implement and stabilize phase 1. Successfully use OOB
 - Need a demo instance, generic catalog, standards, consistent processes
 - SB does not want to introduce risk of untrained developer causing a data leak
- Current business trend – when is it a good fit?
 - Effective with large global engineering org, mature on SN platform
 - Challenge is figuring out when Citizen Dev IS appropriate, rather than when it ISN'T appropriate
 - With professional developers, architects, etc., it's hard enough to build and maintain governance; let alone business people who have jobs that aren't IT
 - If the customer has bad practices on the platform already, need to address that before rolling out program
 - Some developers think no one should develop on SN platform without a CAD
 - Citizen Dev is a good fit for custom apps, not development on the baseline/OOB components of the SN Platform

- Sometimes running/managing the program is more work than just having IT develop the apps
- Best if desire to develop comes from the citizen, rather than a mandate from manager
- Getting set up and establishing governance is the start, but success still depends on getting the right people in the role (training PLUS commitment to the process/governance)
- Good CitDev candidates:
 - Tech-savvy, eager volunteers, properly trained, process owners, business analysts, platform experts, developers in previous role, junior developers
 - Check IT Demands backlog. If there is a particular team or department that has a lot of requests, they might be a candidate group to alert to the coming opportunities of Citizen Development
 - Leading practices: Assign a mentor or coach for new Citizen Developers and pilot program
- Ask the questions, how mature is the intake process and how much capacity does IT have? It might be is easy enough and faster for IT to build. But CD is meant to expedite demands/innovation. If the backlog is long, then the program offers value with enough process and tool protections to minimise debt
- Custom table creation and license impacts certainly complicate CitDev.

Other resources

- [ServiceNow Development Pipeline Management Guide](#) This guide can be used by organizations that are seeking improvements in application development, deployment, and testing
- [Process Automation Designer](#) product site
- [Getting Started with Process Automation Designer – Workflow Automation CoE](#) This is a great summary of PAD, with links to lots of other resources including proposed leading practices

- [Citizen Development Core Skills Micro-Cert](#) on Now Learning
- [Setting yourself up for success with KPI Composer](#)
- [Performance Analytics Academy](#): PA personas - there is a good slide here at 5:39 - "analytics as a team sport"

Weekly Study Guide (Week 9)

Facilitator Notes and Presentation Tips:

- Complex graphics with lots of categories, relationships or phases? Try to simplify and break it down to concepts. Try colour-coding to help do this visually.
- Architecture and integration diagrams can be complex! To help your audience follow along, use the tools of your presentation software:
 - Slide animation features can be used to hide/show features. A quick trick is to have shapes that cover parts of the diagram and then use animation to hide them, allowing the diagram underneath to show through. Another option is to build out the architecture across multiple slides.
 - I have seen numerous examples in your slides where relatively complex diagrams can be simplified with animation and colour.
 - Laser pointer can help focus on the particular part of the diagram you're discussing. Be sure to turn this off again when not needed.
- For more detailed/busy slides, consider a simpler slide for the presentation and add the detailed slide in an appendix. You can leverage the appendix slides if there are questions on the topic
- This week, we didn't really need to cover the importance of foundation data again since this was already covered in the data strategy week. But note that foundation data is often the key parameters for a process. To have reliable processes, quality foundation data is also required.

Integration Leading Practices:

- Prefer discussing leading practices in line with the solution, rather than as a separate list. The additional context helps the audience understand how the leading practices will be applied to their situation.
- Phase 1 of implementation should include foundation data. Examples of systems/sources/methods used in phase 1: active directory, SAML 2.0 idP, Workday, SuccessFactors. Skipping over any of these fundamentals should be a red flag... if your customer suggests doing so you should be ready to push back (say “No”).
 - Refer back to the data strategy content (week 5) for how to decide which data is needed, and when. That can be a guide for planning an integration roadmap for automating the data import.
- Integrations are complex, and usually have long lifecycles – they take significant effort to build, test and maintain. Therefore, it’s important to make good decisions throughout the life of an integration – planning, design, implementation, deployment, operation and retirement/replacement. What’s the framework for making good decisions??? GOVERNANCE! Remember your governance topics from Week 4, and consider how the ServiceNow governance framework would apply to integrations (including all the points below).
- Bi-directional vs uni-directional integrations:
 - Uni-directional most appropriate for ServiceNow consuming data e.g. foundation data feeds
 - Bi-directional most appropriate for more complex process integrations where both sides need to coordinate their respective parts of the process. E.g. e-bonding and SIAM use cases.
 - Additional complexity of bi-directional integrations can be more than 2x a uni-directional example. It’s not simply an integration in each direction... you may also need to solve issues such as race conditions, data ownership and endless looping of updates.
 - Consider push or pull, inbound or outbound, network security implications and potential lead times e.g. firewall configuration changes
 - Use correlation IDs to understand the linkage of data across the integration

- In diagrams, be clear if you have directionality shown with arrows whether that reflects connection direction or data flow direction. Don't assume that the directionality is commonly understood by your audience.
- Foundational data guiding principles: Gather from source of truth, complete/accurate data, minimize overhead, reduce capability onboarding time, and maximize platform usage
 - Include data stewards (org wide) to determine master data source(s). Ensure source is being managed and maintained
 - May need executive stakeholder to weigh in when teams with incompatible data claim to be the source of truth
- ServiceNow is a system of engagement and system of work, not a system of record for foundation data
- Outages and operational support of integrations
 - Implementation can include methods to handle error conditions e.g. message retries, auto-raise an incident if retries fail. IntegrationHub has some built-in error handling capabilities
 - Consider operational needs for supporting integrations: logging and monitoring, error investigation and troubleshooting, retry/replay, reporting and metrics
- Consider remote tables as a substitute to ServiceNow being a dumping ground for data
- Attachments can still be problematic in ServiceNow integrations. Typically the attachment needs to be sent/received separately. There are often limitations when sending/receiving attachments. E.g. ServiceNow doesn't support inbound messages with multi-part format (multiple attachments).
- Architecture diagram is helpful for data and integrations. It's a living document.
- What is the value of building an integration to import/migrate legacy data? Things to consider:
 - Evaluate the quality of legacy data against data governance objectives

- Sometimes legacy systems may be the only source for certain data attributes, especially non-discoverable data, which may increase the priority of migrating this data
- Implementation phases may simplify the situation if Discovery and other automated methods are deployed first
- Data security:
 - Authentication options and connection encryption (e.g. TLS)
 - Data security on each end – is the data held securely and compliant with data security, privacy standards etc.?
 - Use of temporary staging tables within the integration – supports audit of the information passed by the integration. What controls over the data (including final storage AND any staging).

Integrations:

- Options: ServiceNow apps/ plugins, ServiceNow Store, Integration Hub Spokes, Build your own... and now a new one – Robotic Process Automation (RPA).
- Integration and documentation are intertwined – Include the WHY (why did we build when there is a spoke?). Stories used for implementation are not sufficient documentation to justify integration design decisions.
- Enterprise Service Bus is in a future phase due to Sovereign Bank's project to centralize/streamline current ESBs. When using ESB need to be aware of the reliance on other team/owner for issues. Do not want rework or scope creep.
- Security is an important piece of the integration conversation, especially with banks and heavily regulated industries. Include the security architect early and often. Security may determine the method of integration
- Build vs. Buy – When you buy, you are on the vendor's time. And an upgrade may surface issues with functionality or support. When you build, you have to consider maintenance/documentation.
- Performance – even if done during down time, too many integrations can negatively impact performance
- Help your customer to know when to stop configuring integrations. Ask the questions: What are you bringing in? Why are you bringing it in? What is it

being used for? How often does the data change? How is documentation managed?

- Socializing and documenting integration schedules is important too - often customer doesn't understand frequency or timing of updates, etc. and will report "bugs" if something isn't updated immediately
- Sometimes there is only one option to recommend (e.g., for Azure AD SSO and Provisioning the plugin is the only answer). But if there are multiple correct answers, call out the different options with pros/cons and costs/effort/time so the customer can pick what they want, and you can give recommendation

Integration Protocols & Integration Hub:

- Even with all of the spokes available, Integration Hub is not always the answer:
 - Large data volumes
 - Synchronous execution requirement
- Licensing considerations – there may be a buy vs build strategy, but there are important value conversations that need to occur to find the right tier of Integration Hub license
 - Once the IntegrationHub tier is chosen, you can choose to use off-the-shelf spokes that are available to that tier. Changing tier is a business decision rather than purely technical. Understand the role of the governance structure in assisting with this kind of decision
- Sovereign Bank spokes for Teams, Slack, SAP,

MID Servers:

- What is the cost of a MID Server to a customer? (There are no ServiceNow license cost. Comes down to hosting/hardware and OS, and operating costs such as maintenance/support)
- Consider number and location of MID servers based on:
 - Sizing

- Function
- High availability design e.g. MID clusters
- Placement for security and performance considerations
- MID Servers should have Line of Sight to the target systems without having to go through network firewalls.
- MID Servers need access to the internet (direct or via proxy) to reach the ServiceNow instance.
- MID Servers essentially poll the ServiceNow instance every 5-15 seconds looking for work to execute in the customer's environment. In this regard MID Server integrations are not synchronous.
- In some limited use cases the MID Server is the target of data triggered from the customer's environment and passed to the SN instance. An example of this is the MID Server being the target for a triggered network SNMP trap.

Other resources:

TechNow episodes that might be useful:

- Ep. 82 – [Instance Data Replication](#)
- Ep. 76 – [Flow Designer/Integration Hub 2020 Update](#)
- Ep. 60 – [How a MID Server can help connect your data/apps/resources](#)
- Ep. 56 – [Extend Integration Hub with Custom Spokes](#)

Weekly Study Notes (Week 8)

Service Portal GenAI/AI and Virtual Agent

Facilitator Notes / Presentation Tips

- As stated in my feedback, it is beneficial in your presentations to be clear in stating what you are recommending for Sovereign Bank. Try and use the key takeaways section to re-iterate and summarize what you are recommending

for SB. Remember that your audience will be looking for your opinion and recommendations for the challenges that are raised each week.

- Also on a smaller point, as I stated in my feedback last week, try and use nouns and verbs in relation to your agenda items. So for example instead having an agenda item such as “GenAI”, think of using something like “Productivity benefits with GenAI” or “User empowerment with Virtual Agent”. Also, by creating a positive approach as this it will help you focus on the key points that you want to put across to your audience. As such the benefits the technology brings to the customer.

Service Portal

- Common approach for SB: Employee Center and two CSM portals (B2B and B2C) in phase 1
 - Different audiences have different preferences and needs (internal, external, cross-dept)
 - Consider your personas and what they want to do
 - Some audiences will need access to more than one portal
 - What about master/unified portal? Consider security, performance and UX impact (more than 4 clicks). A hybrid approach with the same entrance (like a hotel lobby)
- Keep mobility in mind. If it’s not a requirement today, consider it strategically. There may be network architecture and security decisions that could make mobility difficult (e.g. IP whitelisting, Edge Encryption).
- What do customers need consider for Service Portal design/maintenance?
 - Available widgets (what they are, what they do and how info is populated)
 - Widgets, reusability is very powerful. Just make sure you put it through an SDLC process and lifecycle
 - ServiceNow is a platform and has its own limitations and leading practice
 - Phase 1 is not a website like Netflix or Amazon

- Marketing/branding seeks innovation and flashy
- User adoption is challenging – corporate material isn't typically exciting
- Design effective content before worrying about the portal design
- Start with the end in mind (use metrics)
- Stay up to date on release information
- Security – Keep in mind that penetration tests should be carried out before releasing because you are building something custom
- Making it contextually relevant – Based on where you are at and what you want to do, what should you have available to you?
- A UX strategy is critical to ensuring the customer is on a path towards their outcomes
 - Have a goal
 - Metrics will tell you about how you are going towards meeting your goal
 - Monitoring and Health tools will tell you about your success
 - <https://www.servicenow.com/docs/csh?topicname=user-exp-analytics-landing.html&version=latest>
 -

Virtual Agent

- The whole point of Virtual Agent is to help with Self-Service and defection
 - Password reset is often the most popular use case when paired with automation
 - Dynamic Translation/NLU/Hand-Off to physical agents
 - Training + processes to ensure agents are available. i.e. 24 x 7 follow the sun if a handoff is required
- Roadmap: start with small initiative (top use cases that bring the most value), account for multi-languages, create custom topics as needed

- Not everything needs to be in VA – there are other options like searching a knowledge base
- VA can be especially powerful in established implementations where the client has multiple business units utilizing Service Portal (now days that would be Employee Center). VA can operate as a utility navigation (like Search) to get users to the right catalogue item, Knowledge, or other page across all business units.
- Think of how VA and chat are like other things we are familiar with (and may prefer)
 - Self-checkout at grocery store
 - Using a kiosk or app to order fast food
 - Using an Alexa or Google assistant
- Good first experiences are important for VA adoption. Once the bot experience disappoints it's hard to bring the user back to use it even if you improve it
- Work with L1 support desk to make VA topic decision to drive change. Identify top reasons for creating incidents and then validate whether we can deflect them using Virtual Agent automation
- Need continuous improvement – not a set it and forget it. There will be maintenance costs
- Think about the user experience
 - Have an option to transfer to agent to reduce frustration
 - Integrating with a CTI Bot to get a call back
- Consider how VA interfaces into Portal and PI. There is really only one Virtual Agent, but the conversations can be configured to be available based on context – eg based on the portal.
- Multi-lingual capability is available with Virtual Agent

- I expressed my general opinion with GenAI for code generation. In general, my opinion is that GenAI for scripts is a useful tool to support a skilled developer. For me the developer MUST be able to understand and quality check/explain the code that has been created. Its use would be to generate the start position for the developer to oversee, edit and test. Currently, I would not see GenAI as a replacement for a skilled scripter. But maybe you have a different opinion.
- As we discussed last week, ensure you are aware that ServiceNow AI and GenAI are essentially related but not the same. GenAI in ServiceNow is implemented via series of technical functions under the heading of Now Assist.
- General ServiceNow AI includes applications and functions that include:
 - Search Intelligence
 - Task Intelligence
 - Document Intelligence
 - Analytics Intelligence
- Document Intelligence: <https://www.servicenow.com/docs/csh?topicname=exploring-docintel.html&version=latest>
- AI Search: <https://www.servicenow.com/docs/csh?topicname=overview-ais.html&version=latest>
- GenAI/Now Assist
 - Now Assist for ITSM
 - Now Assist for SPM
 - Now Assist for CSM
 - Now Assist for HRSD
 - Now Assist for Creator

- As discussed Now Assist can provide real productivity benefits with supporting functions such as: Incident Summaries, generating Resolution Notes, creating start position Scripts/Flows, summarize chat interactions and more. See reference: <https://www.servicenow.com/docs/csh?topicname=exploring-now-assist-platform.html&version=latest>
- Now Assist: https://youtu.be/yBX-uVTs_I?si=-6pzkSN7HOe2-CXB

More Resources – Discussion Items

- [Forget about algorithms and models — Learn how to solve problems first](#)
- [What is predictive analytics and how can you use it today?](#)
- [AI and predictive analytics: What does the future hold?](#)
- [5 examples of PI](#)
- [PI Workbench Demo](#) on the Community

Other topics of interest:

- Service Portal: Is it being
Deprecated: <https://www.servicenow.com/community/service-portal-articles/is-service-portal-being-deprecated/ta-p/2505827>
- AngularJS Long-term
support: https://support.servicenow.com/kb?id=kb_article_view&sysparm_article=KB0965345
- Predictive Intelligence
FAQ: https://support.servicenow.com/kb?id=kb_article_view&sysparm_article=KB0781894

Weekly Study Notes (Week 7)

User Experience, Information Architecture, Interface Design

Facilitator Notes and Presentation Tips:

- Goals vs Agenda
 - Consider under what circumstance you use each of these and make it intentional in your presentations
 - Goals can be used to provide the audience a view of what we will do today
 - The Agenda can be used to provide a view of how we will get there
 - Goals can also be used in the structure of a presentation for:
 - Tell them what you will tell them
 - The presentation can be used to “tell them”
 - The Summary can be used to “remind them what you told them”
 - This technique is known as “Tell them, tell them, tell them”
- Think of your audience
 - Keep in mind you have a UX person in the room.
 - How much would they need to know?
 - There is also an Enterprise Architect, what would they need to know?

User Experience:

- The key elements for UI design: Simplification, Standardization and Usability.
- Where/How are our users accessing ServiceNow from?
 - Mobile : NowMobile, NowAgent
 - Portals : CSM B2B, B2C, Employee Center
- Workspaces
 - Role based

- Consolidated view
- Adoption of this has not been easy based on cohort experience
 - OCM is key to this – Stakeholder + Impact analysis
 - A good way of introduction a new UX such as this is by using a Pilot
 - Gather feedback as you go and find out what works/what doesn't work
 - New customers will adopt
 - Existing customers will always have something they compare it too, hence will be resistant
- Analytics are available to help you understand areas where UX is effective. Continuously monitor those to ensure you understand where to invest your time to improve.

UI Design:

- As I mentioned in my Service Catalog example, it is easy to make incorrect assumptions about your users and the process you are trying to digitize.
- In my opinion, you need to develop robust testing processes not just for functionality but for usability. In testing it is also important to test for marginal/extreme inputs as well as the expected inputs. So for example you need for a future date input, you would test for:
 - Representative: Today + 1
 - Marginal: Today
 - Exterme: Today + 200
 - Invalid: Yesterday OR Null or Not a Date

Information Architecture:

- Possible Taxonomy Scheme for Sovereign Bank:
 - Consumer Banking/Private Banking/Commercial Banking

- Accounts/Cards/Loans/Tax
- Change of details/Application for an additional cardholder/Home loan discharge
- Organisational scheme: Geographic and topic schemes could make sense for Sovereign Bank
- Functional scheme: ITSM/HR/CSM/Requests and Tasks
- Majority of businesses use hierarchy to organise the structure. Many users are familiar with the parent/child relationships
- Performance is also an aspect that usually affects UX while not being directly connected with information architecture or schemas
- Impact of Taxonomy
 - Capturing the right data...? For...? Access? Reporting?
- Are the categorisations you pick for knowledge going to be applicable for the Service Catalog?
- Employee Center - organises into Topics already and guidance is to supersede Service Portal
- Governance can play a role in defining policy for how the Information Architecture is defined and how it can change. Governance can also help decide what variations are allowed in UX to meet various persona needs, and the technical approach to meet those needs. UI/UX can be a source of demand to customize, and governance is the process an organization can use to maintain control.

Interface Design:

- People want an Amazon/eBay experience. Is this realistic?
 - The level of research performed by these organisations is huge. i.e. 1-Click checkout
 - People want an Amazon or Google-like experience. But what is that? Ask the probing questions to find out why these examples are given and which parts of those example experiences the customer imagines

in their own implementation? (You might use the 5-Why's technique here)

- Would be good to know who is using the technology before diving into architecture (persona centric view). Structure and scheme depends on the audience. Not a one size fits all. Therefore, ensure to provide the context (which persona groups, performing which tasks or using which ServiceNow applications/capabilities) when presenting recommendations on information architecture.
 - Key personas: Employees – internal requesters, Service Desk Agents, IT and other process users (fulfillers), external customers and consumers. Can also break down by business/org e.g. HR, IT, Finance etc.
 - Key applications/capabilities: Service Portal and Employee Centre, traditional platform UI (forms interface), Workspaces, knowledge base, service catalog.
 - Define who are the users, what are they trying to do?
 - Are they a customer?
 - Employee? IT, HR?
 - Walk a mile in the end user's shoes
 - Maturity of the users matters. Some are happy with the classic view, otherwise want something like workspace
- Form replication of the service catalog
 - We often tell customers not to replicate paper forms on the service catalog, but in a highly regulated industry like banking, is this feasible?
 - Sometimes things need to be replicated due to regulation, i.e. capturing specific fields that may seem redundant
- Search - Reclassification and review of KB Data
 - AI & Automation can help in information organisation and searching
- Surveys are a good way of gauging feedback
- What tools are available OOTB to help us with navigation? Guided Tours

- Accessibility is a key consideration when designing interface design. The Web Content Accessibility Guideline (WCAG) has been defined by W3C to help this (<https://www.w3.org/WAI/standards-guidelines/wcag/>)
 - Accessibility can be for visually impaired (using screen readers)
 - Language is a part of UX
 - Multi-Language for
 - Service Portal
 - Catalog Items or Knowledge Bases
 - Using gs.message to get translation based on the user's profile
 - Not a good experience - some specific things get translated, but not others
 - A “laptop” or “desktop” may be a part of the language an IT group has, but for an end-user in the business it might just be a “Personal Computer”
 - Using impacts instead of words helps break this down
- Re-using components is important
 - Leads to less work
 - Standardisation of capability and user experience

Multi-Language Interfaces:

- ServiceNow provides a lot of support for organisations to implement multi-language UIs and Content. ServiceNow offers something called the Localisation Framework:
 - <https://www.servicenow.com/docs/csh?topicname=localization-framework-landing.html&version=latest>
- You may want to watch this helpful video that explores the Localisation Framework:
 - <https://www.youtube.com/watch?v=IMlx40Jnfvo>

Some Helpful Resources:

- The [Ideas portal](https://support.servicenow.com/ideas) (<https://support.servicenow.com/ideas>) on the community can be used to provide feedback/enhancement ideas to the ServiceNow product manager
- Organisational Change Management / aka Adoption – “What value is there in building something if people don’t use it?”
 - Prosci is an organisation that have defined a framework for OCM - <https://www.prosci.com/resources/articles/prosci-methodology>
- The “5 Whys” is a technique that can be used to identify the driving need that someone has as opposed to the solution they've put forward
 - Watch this video for an example of this - <https://youtu.be/BEQvq99PZwo>
- [Design Process: From Chaos to Order](#)
- [The Nine Principles of UX Design Psychology: Can you Predict the Behavior of Users?](#)
- Matt Abrahams – communication coach www.mattabrahams.com
- Guided Tours with Next Experience - <https://www.servicenow.com/community/admin-experience-articles/using-guided-tours-with-next-experience-ui/ta-p/2420238>
- WCAG Guidelines - <https://www.servicenow.com/au/accessibility-statement.html>
- AI Search - <https://www.servicenow.com/products/ai-search.html>
- Next Experience architecture and terminology <https://www.servicenow.com/community/next-experience-articles/next-experience-architecture-and-terminology/ta-p/2331947>

Weekly Study Notes (Week 6)

Security, Encryption and Scoped Applications

Facilitator Notes and Presentation Tips

- Be careful in your weekly group work to avoid just explaining the online learning for the week. Please focus on the *application* of the content to the case study you're given, with specific attention to the outcomes that Sovereign Bank wants to achieve. Spending too much time explaining the theory leaves you with not enough time to justify how your recommendation will achieve desired results – this is important always, and especially critical for a 10 minute presentation.
- - Make recommendations. Guide the customer based on what you have learnt or have experienced
 - Benefits will only be of value if they're specific to the customer. Generic benefits might help a product sell cycle, but when planning an implementation, think about what the customer gets practically
- Augment your learning with other resources beyond the online learning from Intrepid.
- Managing nerves in presentations:
- - Think how you can control your breathing... breathing slower and trying to be physically calm can help you relax emotionally as well.
 - Lot's of people find that their nerves subside as they get further into the presentation – this often comes because focus has shifted to talking about things that you know about and away from anxiety-causing questions about how you might be perceived.
 - Some positive self-talk before your presentation can remind you that you're ready. You got this!
 - Learn to recognise what you do when you're nervous and have a strategy to counteract. E.g. if you speak too quickly when you're nervous, make yourself go slower. If you fidget or rock side to side, tell yourself to be still.
 - Think how long it might take someone to understand your message, and try not to go any faster than that. If you take a small pause before you begin explaining a point, it can help get your thoughts in order AND can help your audience understand.

Security Architecture

- Recommend the Instance Security Center to harden instance security and receive daily compliance score instance security alerts. Note : Instance Security Center is being prepared for future deprecation. It will be hidden and no longer activated on new instances but will continue to be supported. ServiceNow Security Center provides the latest experience for this functionality. <https://www.servicenow.com/docs/bundle/yokohama-platform-security/page/administer/security-center/concept/sec-center-v2.html>
- You can use combinations of solutions. Some examples:
 - Edge and Column/Platform Level encryption
 - Start with Platform Encryption with DB Encryption, and do a PoC to see if Edge Encryption is needed.
- Layered approaches to security can be beneficial. E.g.
 - Edge Encryption for data sovereignty
 - Column-level encryption to augment ACLs for access to field data
 - Database encryption for compliance with policy for encryption-at-rest
 - Full Disk Encryption to mitigate risks for data centre breaches
- Not all data security challenges need to be solved with encryption. Consider all the platform capabilities such as ACLs, role-based access controls, adaptive authentication (and others)
- When evaluating different encryption options, the customer should take a risk-based approach. What is the specific risk they are concerned about (in line with policy) and what are the options to mitigate?
 - Mitigations don't all have to be technology e.g. ACLs and encryption. They can also be focused on people and process e.g. audits on admin accounts and an approval process for all new admins.
- Organisations often will define their own policies for application and data security. These policies will generally take into account any external regulatory requirements. The easy path for your ServiceNow solution design is to be compliant with the relevant policies. However, it's common for security policies to be very conservative and prescribe certain approaches to

mitigate risks. If this leads to complex/difficult solutions in ServiceNow, work with security architects to understand the risks being controlled and consider all options (not just technical/encryption) to manage the risk.

- The [5 Whys](#) technique can be very helpful here to truly understand what's important.
- Edge Encryption (Keep in mind, this is going away, see other posts in forum.):
 - Use-cases: sensitive data that can never leave customer network and data that cannot leave a country/region
 - Need to articulate the additional complexity/cost/management associated with Edge Encryption
 - ServiceNow does not have access to data encrypted by Edge, so consider implications for reporting and any server-side workflow
 - What if proxy servers are unavailable? Encrypted fields will become inaccessible
 - For Sovereign Bank, teams that recommended Edge Encryption cited Tokenization / Pattern matching as a key benefit
- Column-Level encryption vs ACLs:
 - This is normally one of those 'better together' scenarios
 - Both can be used to control access at the field level, but only ACLs control access at the record level
 - ACLs control access, but don't encrypt the data, so CLE/CLEE is needed if encryption is a must.
 - Creating an encryption context and encrypting a field can be done independently of the ACL design, so can be a good way to add data controls without impacting OOB ACLs
 - You can define encryption based on location using custom scripts or by defining roles based on location
- Database Encryption is a software-based encryption mechanism. Full Disk Encryption is hardware-based with dedicated physical disks being used.

- There is a scalability risk to the customer with Full Disk Encryption as the hardware disk unit can only support 'x' instances before the space is consumed.
- Useful discussion questions:
 - How to secure the Network Layer?
 - IP restrictions is a bit of a blunt instrument.
 - Using Adaptive Authentication provides a much more granular approach to restricting access to the ServiceNow instance. Thus functionality to set up to restrict certain users based upon a white-list of IP addresses and also allow remote workers to access via registered devices. <https://www.servicenow.com/docs/bundle/yokohama-platform-security/page/integrate/authentication/concept/adaptive-authentication.html>
 - There is also a relatively new feature called Zero Trust Access which can restrict user permissions based upon the network that they are using to access the instance. Thus an admin may have restricted right as they login, if they login from a public network. <https://www.servicenow.com/docs/bundle/yokohama-platform-security/page/integrate/authentication/concept/explore-zero-trust-access.html>
 - Keep in mind Customers use VPNs to allow access to their corporate resources. Access to ServiceNow could be permitted through this also
- Data Sovereignty
 - A podcast on Data Sovereignty: <https://www.servicenow.com/community/nw-platform-blog/the-enterprise-cloudcast-episode-9-data-sovereignty-by-allan/ba-p/2291469>

- Update sets continue to be relevant, but don't ignore the newer features and mechanisms for managing and migrating configuration. When using update sets, batching them has been a common practice. Other platform features such as source integration and Application Repository mean that update sets are not always the best approach for code migration/deployment. So... what to use?
 - One approach is that it might depend on the amount of changes. For more significant changes, create scoped apps as containers and use the App Repository for migration and deployment. Use update sets for small/minimal changes to OOB.
- Why use the application repository?
 - Multiple developers can work on the app and not worry about update sets
 - Safe installation process in upper instances once you have tested the promotion
 - Ability to downgrade to previous version in upper instances should the need arise
 - Installing and updating applications on all company instances without the need to manage update sets or local files
- A benefit of App Repository is that it should be simple to detach and roll back to OOB if ServiceNow includes the required capability in a new release.
- If you started app development using the App Repository, continue to use it. Don't combine both update sets and App Repository for scoped app development – it can lead to issues with deployments and rollbacks.
- On-prem customers don't have access to App Repository
- When using App Repository, the process for creating new app versions should be controlled to avoid untested code being included. This is especially relevant for larger teams.
 - One approach to address this risk is to combine Source (GIT) Integration and App Repository. Use the Source Integration feature to migrate code from DEV->TEST. Once testing passes in TEST environment, then create a new app version from TEST and use App

Repository to deploy to later environments. This keeps untested code in the DEV environment from becoming part of an app version.

Some Helpful Resources:

- [5 Whys technique](#) for getting to the root of a problem
- [Safeguarding Your Data](#) – An overview of data security
- [Cloud Encryption with Key Management](#) product docs
- [Now Support](#): How to go back to using app repo after committing changes to an application via update sets
- [Community Blog - Encrypting columns \(fields\) using "Column Level Encryption"](#)
- [Trust Compliance Center site](#) – Search here, there are answers regarding who can access the servicenow datacenter , good to answer the tricky question
- A whitepaper on app development from one of our employees in Germany: [A mature Development and Deployment Process](#)
- [The Custom Table Whitepaper](#)
- Developing in the Global
Scope: <https://www.youtube.com/watch?v=ttBKE8w05cs>
- Discussion over GiT, Update Sets and App
repo: <https://www.servicenow.com/community/now-platform-articles/a-simple-deploy-strategy-involving-git-repository-for-custom/ta-p/2316029>

Weekly Study Notes (Week 5)

Facilitator Notes and Presentation Tips

- It's a great idea to watch the recording of the other VCS. As much as we try to steer the conversation based on curriculum and learning objectives, it's common to see different discussion topics across the two sessions, and even when the topics seem similar, we can also see some different perspectives.

- It's important for your presentations to be cohesive – there should be a structure that makes it easy for your audience to understand how things will fit together to meet their requirements. See the Presentation Skills tips for some more on this.
- When you make a recommendation, be prepared to share why you are recommending it (be prepared to defend your recommendations) and what are the considerations or risks that we might face always try to tie these back to the outcomes and challenges of Sovereign Bank.
- Solution designs are greatly influenced by the design inputs / parameters / constraints and their relative priorities. If you explain how, you have considered these inputs in arriving at your solution recommendation (e.g. by stating your assumptions) then you leave yourself room to reconsider the solution design if your customer prioritizes things differently to you.
- When you consider a recommended solution, sometimes the ideal is not immediately achievable. Remember that your decision is in the context of Sovereign Bank which has had historical issues and current strengths and weaknesses. A solution must have a level of practicality for the customer.

Data Strategies

- In many organisations, data stewards/owners work in silos and can be very possessive of their data. This is important to keep in mind when bringing their data into ServiceNow where there may be a platform approach to data, and stronger governance over how data is managed – they may not have the same amount of control over their data as before.
- Data stewards/owners should focus on data quality. They may not have direct influence over processes and how data is used, but accurate and complete data can improve the effectiveness of many processes.
- There is not a one go-to data architecture – go with the architecture that meets the business outcomes.

- Legal / CISO can completely torpedo any solution – understand the business outcomes and be ready to defend your solution and review alternatives. Compliance with policies (internal, external or regulatory) that seek to keep data secure is very important.
- Legacy data aligns to the legacy process, and if migrated to ServiceNow can create resistance to process change/transformation. A good architecture is to get the customer to migrate legacy (closed) transactional data to a data lake/warehouse for reporting and audit purposes and either migrate only the open records or do a dual run to help close out active records in the legacy system and create new in ServiceNow. This separates legacy data and process from the new (and hopefully improved) version.
- If there is push back on loading data, then ask the question for what the value of migrating the data is. What do they want to achieve? (Use the 7 Why's).
- Auditing requirements can often drive how old data needs to be accessed/reviewed. Include auditing early in those decisions to help make decisions on 'old data'. Per the point above, don't assume data MUST reside in ServiceNow as it may be possible to run reports from an archive database or data lake.
- There is a danger of 'set and forget' behavior with data governance, which can lead to increased risk as time passes and things change/evolve. Address this with continuous monitoring and review of governance processes. The concept of continuous improvement can be powerful here when applied both to data strategy AND to the governance processes that oversee the data strategy.
- Because of the effort involved in managing the correctness and compliance of data in ServiceNow, it's important to only import data that is going to be used directly by processes hosted in ServiceNow. This applies to both the entity and attribute layers.
- For Integrations, Enterprise customers tend to formalise architectural 'patterns' which is a solution to a specific problem. These can start with standard adapters and connectors which would lower the technical debt. For this, we can firstly look to IntegrationHUB. Store based connectors, 3rd party connectors and custom build of integrations can then be considered.
- IntegrationHUB licensing is based on volumes of transactions, Keep this in mind when using it.

Instance Data Replication

- IDR is- ServiceNow to ServiceNow only.
- The decision to go with multiple production instances comes *before* a decision to use IDR. IDR then becomes a mechanism to keep certain data synchronized across those instances.
- If the data to be synchronized is small and/or simple to manage, there may be other options:
- Integrating each instance to the same source data system (e.g., AD for user data).
- Small point-to-point integrations between the instances that use other technologies e.g, eBonding or web services.

Domain Separation

Why Single instance for Sovereign Bank?

- Single instance is the default position that provides the simplest architecture for consolidating data and process. Other more advanced architectures such as multi-instance or domain-separation may be considered if the single-instance architecture is inadequate.
- Single instance reduces the need to synchronize common data across instances. This can be a very difficult challenge for large, in flux datasets such as CMDB, or require significantly increased integration effort.

Why Multi-instance for Sovereign Bank? (vs Domain Separation)

- Requirement for GDPR and UK data residency. If everything housed in the UK, would introduce a lot of network latency for internal users, which might hamper adoption.

- Potential new data residency requirements could follow and multi-instance gives flexibility to host instances in other jurisdictions.
- Sovereign Bank desired business outcome is for a unified process.
- Can use data replication with restrictions on PII and HR data.
- ACLs, security big reason for the logical separation.
- Domain Separation is irreversible and has high maintenance overhead.

Why Domain Separation for Sovereign Bank?

- Different areas of the world with different rules
- Want to avoid heavy customizations for security rules
- Separate customer and HR data
- Consistent process applied at the required levels but the ability to modify in sub domains as required.
- Good where the tenant has limited process variations
- Faster adoption/onboarding

Additional considerations for Domain Separation

- Deciding for/against Domain Separation is greatly influenced by whether the business strategy/direction is aligned/not aligned with the product intent behind the Domain Separation product (i.e. Managed Service Provider (MSP) use cases). Misalignment is likely to result in requirements that are difficult to meet without significant risk.
- Need to prove value of process separation - need to research this option extensively. For Sovereign Bank, their desire to consolidate process, and to simplify and automate, seem to be at odds with the idea of segregating process with Domain Separation. Therefore, use caution and be sure any decision is made with long-term value in mind.
- Lots of overhead required to manage a domain-separated instance. General BAU administration and development is different compared to non-domain-separated instances. All administrators and developers need to be enabled on the additional requirements for Domain Separation.

Hoops to jump through to get anything accomplished in a domain separated environment is a roadblock for delivering a valuable product.

- Risk of data exposure via administrator error in Domain Separation (compliance risk)

Some Helpful Resources:

- “ServiceNow 'schema' and 'meta' APIs, and why system integrations should use them” - <https://davidmac.pro/posts/2021-11-26-schema-meta-api/>
- Excluded tables from IDR: <https://www.servicenow.com/docs/bundle/yokohama-servicenow-platform/page/administer/instance-data-replication/reference/IDR-excluded-tables.html>
- <https://youtu.be/6V30le6hbw8> - Youtube video “Domain separation overview for service providers”. Note the following quote from early in this video... *“Domain separation is intended for specific use cases. The primary one is for service providers who provide managed services where it’s important to separate their customers’ data.”*
- <https://youtu.be/C7lesKlzyig> - Youtube video “TechNow Ep 82 | Exchange data between ServiceNow instances with IDR”
- Domain Separation:
 - Domain Separation Basics course in Now Learning
 - Domain Separation, part 1: Do I need it? What are the alternative options?
 - Domain Separation Workshop pack on NowCreate
- Download this comprehensive reference to ServiceNow knowledge resources organised by topic by Maik Skoddow: <https://www.servicenow.com/community/training-and-certifications/knowledge-sources-to-go/ta-p/2310130>
- Know your privacy laws: Data protection and privacy legislation worldwide: <https://unctad.org/page/data-protection-and-privacy-legislation-worldwide>
- Integrating to/from ServiceNow instances:

<https://www.servicenow.com/community/now-platform-articles/data-integrations-from-and-to-servicenowinstances-part-1/ta-p/2308545>

- Post-Production Domain Separation Activation

Utility: <https://www.servicenow.com/docs/bundle/yokohama-platform-security/page/administer/company-and-domain-separation/concept/post-prod-domain-sep-utility.html>

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View 1 ReplyReply

- Instructor

Weekly Study Notes (Week 4)

Agile Frameworks, Technical Governance, Release and Instance Strategy

Agile Frameworks

- Scrum of Scrums vs SAFe
- One key difference that we see being used to decide this is whether the organisation has a Portfolio Manager to allow for additional oversight of what's being developed. SAFe caters to this model well.
- Keep in mind that there is a level of maturity required to implement SAFe. You could find that starting with Scrum of Scrums and then transitioning to SAFe might be appropriate, depending on where the customer is at. It is very situational for how you could introduce this. Remember SAFe requires multiple Scrum Teams with developers. Consider so far SB have only really been developing on the HR instance and has had problems.
- Why SAFe
- It has the Release Train methodology

- SAFe needs to have more OCM, hence using Scrum of Scrums for the implementation and the following first phases of releases may be considered.
- Yes it may be tough to start to make team to SAFe from scrum , but it will be proven factor once we start with at least essential SAFe
- Maybe Start with Scrum Of Scrum and then transition into SAFe?
- Which Safe, what Trains would be needed? Maybe Platform, HR, ITSM, ITOM, CSM?
- Is the CTA going to decide the strategy on whether Scrum of Scrums or SAFe is used?
- The CTA would have a hand in guiding the customer in the most suitable methodology to use based on the customer's circumstance/context and strategic goal
- The customer may also already have an implementation methodology that they want to adopt
- The key here is for the CTA to understand what the key considerations would be in deciding between SoS vs SAFe so that they can talk to this with a customer

Technical Governance

- Foundation and Data Quality - flows into Technical Governance - Data Architecture
- Importance of Golden Rules
- "Just because you can, doesn't mean you should"
- Interrelationships between governance boards
- Think of examples that can be used to demonstrate this
- How does the governance regime handle conflict?
- The Business Smart customisation process is a critical one that needs to be understood as part of the way that we evaluate requirements in terms of value vs risk to ensure that we are aware of and actively accepting these

items (which constitute technical debt). This is a key activity of the Technical Governance Board. This is a good way to start with the conversation for the smaller customers

- This pack from Customer Success does a good job in outlining this - https://learning.servicenow.com/nowcreate/en/pages/assets?asset_id=a820de08c3a30a501ac0f60f05013157&id=nc_asset&nc_source=copy_asset_link
- Defining metrics for technical governance at the start is an important aspect as this measures where we are at and whether we have achieved our goals and also demonstrating the value of running the process
- How do we apply the governance structure recommended to smaller organisations?
- You don't need to prescriptively stand up all of the 'boards' formally, as long as there is an ability for the same outcomes to be achieved. Governance can be there without the formality of the boards.
- You may consider changing the term 'boards' to 'meetings' to make it more conducive to being made smaller
- Starting with at least Technical Governance and demand management is the minimum you'd want to start

Instance and Release Strategy

- An Instance Strategy supports a Release process. Think about how you manage your SDLC, including Testing (different variations on testing) and how to 'release' products into Production
- Factors determining strategy
- Experience
- Size of the customer
- Size of support capability
- Regulatory requirements
- Size of implementation

- Multiple Production instances – what are the drivers that would result in this? Data? Customer vs Employee? As we will see in week 5 the synchronisation of high volume/high flux data (e.g. CMDB) is not a great strategy.
- Upgrades – when do customers upgrade? once a year or twice a year
- Servicenow contractually needs customer to maintain n-1. Hence a customer would need to upgrade once a year to maintain 'n-1'.
- Operational upgrades - minor/patch - security etc.. released quarterly, then monthly, then on demand
- Upgrade depends on maturity level and magnitude of customizations they made over the years on OOTB applications
- Many teams chose a 4/5 instance structure with Dev, Test, UAT, Prod and Stage/Prod Support with a justification for why each environment exists
- Why additional instances?
- Validation/stability based on the testing that you need to conduct to reduce the risk of a new 'baseline' version having an adverse effect on the business
- A good example would be for validating for migrations
- Mitigation of a risk going wrong in Production, hence a Production Support - Exists so that you have a replica of Production that you can investigate and fix critical defects. Note that anything done through this Production Support pathway would need to also be replicated to the non-production instances
- The production support could also be used as a 'stage' environment before deploying a package in production, then you have somewhere to test the deployment
- Sandpit - play around with capabilities or new features (such as a major release)
- Not having the additional instances will mean that there will be risks and manual processes that will be used (e.g. if you need to upgrade mid-transformation, then you'll need to pause your development to carry out the upgrade).
- Remember however many instances you choose that must all have a tight security policy so as not to allow weaknesses or vulnerabilities.

- What influences how many Dev instances you need?
- One Dev instance works well if there are guidelines
 - If the transformation that you're involved with involve many products, which will mean that you will have different application scopes, then the risk of conflict is reduced
 - How the different groups communicate with each other and align with regards to working on the same objects etc...
 - Using a "Technical Scrum" helps here which involves bringing together leads from each scrum team to a forum with the platform owner + architect so that that each lead can talk about what their scrum is focussed on for the week. The Key factor here is around communication and ensuring that all devs are continuously talking about what they are doing to help avoid conflicts
- Parallel releases
 - Doing development on the same product in parallel for different releases
 - How would you manage code synch?
 - Using gitHUB would help in doing this
 - The Release management processes followed would need to be aligned to enable this
 - Multiple Dev instances would need to be used to manage the parallel application development
- Regular Cloning is something that is priceless in managing the instances. The decisions around how often depends on the environment and what it's being used for (Prod -> UAT, more regular whereas Prod -> Dev, not as regular - due to amount of activity that occurs)
- Be aware that there may be some data masking required when cloning data from Production to non-production instances.
- The cloning strategy may come into the conflict with a program office where the cloning has been executed poorly in the past. Hence, it's important to have strong publishing approaches and responsibilities for developers such that communication occurs early on to ensure that everyone is aware of what they need to do

Some Presentation Skills Tips

- Remember the key quote “You are there to solution, not teach”.
- Your presentations need to be in the context of the customer case study. For example, What will Governance address for Sovereign Bank? How does it align to the existing challenges or the key outcomes that Sovereign Bank want to achieve?
- A new tab has opened up for “presentation skills” on Intrepid. Look out for the “Nancy Duarte” videos on presenting
- If you feel that you lack confidence in speaking, lookout for a ‘toastmasters’ club near you. This is an organisation that unites people with a common goal to be able to present and talk comfortably and with confidence. They have great mentoring and club events/activities. <https://www.toastmasters.org/>
- Lastly, someone in a previous cohort shared this video on presenting <https://www.youtube.com/watch?v=soZBrFglqtc>

Some Helpful Links:

- "Release Deployment Process and Operational Best Practice Guide" link https://learning.servicenow.com/nowcreate/en/pages/assets?asset_id=5d95bd8287c4f450ed3b74c9cebb35bc&id=nc_asset
- A Simple Deploy Strategy involving Git Repository for Custom Apps in ServiceNow: <https://www.servicenow.com/community/now-platform-articles/a-simple-deploy-strategy-involving-git-repository-for-custom/ta-p/2316029>
- Now Collaborate with your team in real-time with Developer Sandboxes and Flow as a Code - Xanadu: <https://www.servicenow.com/community/now-platform-blog/now-collaborate-with-your-team-in-real-time-with-developer/ba-p/2935823>
- Source Control Stashing in Tokyo - How to deploy "ready" code to App Repo: <https://www.youtube.com/watch?v=Pi1nyt3jS2Y>
- Manage customizations to applications: <https://docs.servicenow.com/bundle/xanadu-application->

[development/page/build/applications/task/manage-customizations-store-apps.html](#)

- Using Global Scope Apps to make Customizations in Global Scope easier: <https://www.servicenow.com/community/in-other-news/using-global-scope-apps-to-make-customizations-in-global-scope/ba-p/2270861>

Weekly Study Notes (Week 3)

Architecture Standards, Blueprints and current vs 'to-be' architecture

Architecture Standards

- The key takeaway regarding is that there are frameworks out there that customers have in place to represent and model their organisations. The IT4IT model is one that ServiceNow most closely aligns with, but that doesn't mean that a mapping cannot be used especially where the customer has embedded a particular standard throughout their organisation. The key lesson for this though is that the ServiceNow capability model can be aligned with architectural reference models too so that an enterprise architect can see where ServiceNow can map into the framework and the organization.
- If you are going to describe an architectural model as part of your presentations, ensure that the relevance to the customer is explicit. Why should they care? What's your recommendation for what they are doing and why?
- Why do we have standards? So that we have a structure of repeatable methods for getting something done. Architects often have a concept of a 'pattern' to help facilitate these. Any solutions can be patternised so that they are well understood. (i.e. think about an ESB and what kind of integration 'pattern' it provides)

Architecture Blueprints

- What worked well in the presentations
- The ServiceNow capability map is a complex diagram. Always ask yourself, what's the value in showing everything, what do I want the audience to see

and take away from it? Is it how much of the ServiceNow capability they are using? Or which specifically?

- Consider ‘focussing’ on the elements in scope only as part of your discussion. We saw good examples of this with one group doing another diagram purely on just elements in scope and another group using animation and colour to highlight specific blocks to support the approach they were talking through
- Key considerations for Phasing
- It depends on many things including the “MVP” that you’re trying to get to.
- The customer’s key priorities and desires. Think about relating this back to the Desired Business Outcomes and/or strategic drivers. It’s great to talk about when specific products will be deployed, but what will this enable for the business?
- Think about the Agile methodology approach of delivering value sooner. What is the ‘low hanging fruit’?
- What are the key components that have dependencies whereby a component would need to be first?
- What is something that would take the longest to work on/achieve, hence does that determine when it should be in the sequence?
- When would you implement reporting and analytics? Keep in mind that the customer has outcomes they are trying to meet. Metrics can be determined to demonstrate how far along they are in meeting these outcomes. So, reporting and analytics (in some level) is worthwhile looking and considering upfront. Maybe with a roadmap for it maturing over the course of the transformation.
- What is delivered at each phase and why does the phase order make sense?

Current and to-be architecture

- Acknowledgement of HR instance in the current state is important. Keep in mind the transformation is from multiple legacy platforms used for different purposes.
- Foundation Data is a key component here that should be migrated

- Security & Regulatory Compliance
- What considerations could exist around the architecture that would be driven by regulatory compliance?
- What are the regulatory frameworks that the solution needs to comply with? GDPR is one of the strictest. Is it possible to have one single organizational policy to meet European, US and Asian data and privacy regulations?
- Establishing personas and processes and determining a 'security profile' based on these for what data needs to be accessed
- Identifying and classifying data
- Once the data is understood, what are the key compliance requirements and against which data?
- What are the risks that are need to be mitigated (solved against)?
- Who at the customer would know about any existing policies the customer may have to deal with compliance? And how do the 'standard ways' that they deal with compliance match to the capabilities that the ServiceNow platform has?
- Security architects
- Compliance officers
- Solutions may involve:
 - Data sovereignty – Ensuring data for a specific country doesn't leave that country
 - Security mechanisms/features – ACLs, Query Business Rules, Application Scoping etc...
 - Encryption (at rest and in transit)
 - Storing passwords in vaults
 - Data anonymisation when needing to clone to lower environments
 - Data retention and disposal (do we store forever in ServiceNow? Does the customer have a data warehouse available?)
- When talking about architecture, think about the challenges raised

- Talk about solutions you're proposing - the "What", but also the "Why"
- What outcomes are achieved or what challenges are resolved?

Some Presentation Skills Tips

- To make your presentations most relevant to the stakeholders, try to include specific details from the customer circumstance. These will be details from the case study – both the main case study as well as the weekly case study.
- Practicing is the best way to ensure you can deliver your presentation within the time limit. Use the Presentation Skills module in Intrepid and refer back to it often as you practice your presentation craft.
- Some slides are going to be complex in nature and have a lot of information to talk through. Think about whether the amount of content on the slide is going to be distracting the audience from your narrative. If it is, think about techniques you can use to direct the audience to only focus on what you are talking about. Using a mouse pointer can do this. Animation is another way of doing it (animate what you want to talk about so that items display in order or cover the other items and animate the removal of the covering shape)
- Remember what a presentation is as a communication form as compared to a written document. Presentations have a significant rhetorical purpose to invoke a mood/emotion with the audience (e.g. to persuade, inspire) as well as a need to communicate information.

Some Helpful Links:

- If you need some context on GDPR -
<https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/white-paper/wp-preparing-for-the-gdpr.pdf>
- The “Selective Attention Test” -
<https://www.youtube.com/watch?v=vJG698U2Mvo>

- ServiceNow Overview and Interactive Capability Map (POST-SKO 2025 Edition): <https://partnersuccess.servicenow.com/search-result.html?q=servicenow-overview-and-interactive-capability-map&prev=&p=1> (If not possible login, reach out to me by email and I'll fix it for you)

Weekly Study Notes (Week 2)

Articulating the Value of the Platform, Running Workshops & CMDB/CSDM

Want to know what happened in the other Virtual Classroom Session? Watch the recording to get the other discussions on CMDB and CSDM.

Planning Your Presentation

- When approaching any presentation always consider who will be the audience. If you look at the use case scenario, Week 2 presentation is to be presented to the **Configuration Manager**.
- The Configuration Manager is a new role, and is a key person in the audience. From their point of view, there are a few things to keep in mind:
 - The Configuration Manager is likely to already understand what a CMDB is, but probably not what CSDM is. So when explaining/defining concepts (the **what**), focus less on CMDB and more CSDM. How do you show the value that CSDM will bring both to the bank and the Configuration Manager? Linking to the 'challenges' and 'outcomes' continuously is a key way to give you the demonstrative value.
 - There should be a focus on value from the Configuration Manager's perspective, including **how** the proposed solution can achieve it.
 - Be careful of offending any stakeholder who may be attached to some aspect of the current state that you're proposing to change. Beware making value statements or putting down the current situation e.g. calling it 'bad'. Focus more on the positive (value) aspects of the future solution

- Be careful when using statistics to describe expected value of solutions that it doesn't come across as a promise. A good strategy might be to describe similar transformation projects you've worked on in the past, and quote statistics from the benefits those projects achieved. This strategy helps to bolster your own trust with the customer (you've succeeded with previous customers), and paints a picture of the benefits they might realise, without it being a promise. Also if you do use any key stats or a quote, remember to cite the source.
- Also remember to address all of the points in the Case Study. This week the presentation needed to include: (i) Definition, Purpose and Value of the CMDB/CDM, (ii) Identify the stakeholders for the CMDB/CSDM, (iii) Define changes needed for a successful implementation, (iv) Define an implementation roadmap up to the Fly Stage. Some presentations addressed all of these issues and some maybe 3 out of the 4.

CSDM / CMDB

- CSDM allows for different pathways through the crawl/walk/run/fly stages. Depending on need, some customers may choose a different ordering of how they build out their CMDB e.g. Applications first or Service Offerings first. Keep in mind the phases are not prescriptive. Sometimes a maturity model can be more focused towards the key outcomes (by priority) which then tell you about which components of the model to use when. The CMDB is maintained through clear processes and these processes give you tangible business outcomes
- You can introduce the purpose and value of CSDM quite succinctly with a few key points:
 - CSDM is ServiceNow's prescriptive guidance on modelling within the CMDB to Services and business elements.
 - ServiceNow products align to the modelling approach prescribed by CSDM
 - Therefore, you can achieve more value from your ServiceNow investment in a shorter timeframe by adopting the CSDM approach
- CSDM has been through several iterations – version 4 is the current one.

- CSDM concepts are hard to understand when described in abstract. Therefore, examples can be very valuable. Try to show a IT Service as it is represented at the different stages of the CSDM.
- Some useful info from prior cohorts on CMDB population that wasn't discussed in detail in our sessions this week:
 - Discovery and Service Mapping may be the initial methods used, but shouldn't necessarily be the only method used as long as we get the up-to date data constantly of what's in our environment (including relationships)
 - Security / Compliance teams can often be helpful for this too. E.g. every endpoint will generally be required to have security software on it, antivirus, management tools, or maybe be seen by vulnerability scanners which can help you identify the completeness and accuracy of discovery or SG integrations.
 - Data might reside in other 3rd party systems which would be of value to import also using other methods such as our Service Graph connectors or (store-based connectors)
 - Governance of data is critical (including data certification) as part of getting to a good CMDB.
 - Change Management are is also a key process to control the data going into and being updated in the CMDB.
 - Non-Discoverable data does exist also. Support Groups, Criticality, Location may not be able to be discovered, so ensure that there are methods to source these and assure correctness (Data Certification)
 - Key message: there is no one size fits all. What should be considered are different 'patterns' that get authorized and a priority/preference order

Running Workshops

- Keep stakeholders in mind when planning workshops. Each workshop is likely to have a different purpose, and will need different stakeholders to be engaged. A good strategy when planning a series of workshops is to keep the workshop purpose and required stakeholders in mind.

- Here's an example: Some organisations run different workshops focused on "discovery" (explore and educate on product features with demos and decide high-level scope) and then deeper-dive "requirements" workshops (make decisions on implementation actions). These workshops have different purposes, therefore potentially different stakeholders.
- Keep the attention span of workshop attendees in mind... full-day workshops are rarely successful!

Some Presentation Skills Tips

- ServiceNow provides some resources such as [Now Create](#) and the ServiceNow [Customer Success Centre](#). Feel free to use these as you prepare your presentations – it's less work for you!
- Don't forget to address the Sovereign Bank stakeholders directly in your presentations. It doesn't have to be a lot, and it can really help get your audience fully engaged with your message. Try addressing stakeholders by name when you're making a point that is particularly relevant to them. This is relevant for your final capstone review, and the weekly presentations represent a safe space for you to practice.
- To make your presentations most relevant to the stakeholders, try to include specific details from the customer circumstance. These will be details from the case study – both the main case study as well as the weekly case study.
- 10 minutes is a short time, we know! This is intentionally designed so that you practice being succinct and focus on the main message you want your audience to walk away with. It's ok to make the main point and leave the detail for later (e.g. "we'll cover the details in a future session").
- A good wrap-up strategy is to reinforce the key points from the presentation as 'key takeaways', and to finish on a positive, action-oriented note with 'next steps'. Including your key messages in both the introduction and the wrap-up reinforces the importance and makes it more memorable for the audience.
- The VCS discussion this week didn't cover a lot of the 'Running Workshops' or 'Articulating the Value of the Platform' learning modules. Even though we

didn't cover them in VCS, the content there is still valuable. Try and use some of the concepts from those modules in future presentations.

- Practicing is the best way to ensure you can deliver your presentation within the time limit. Use the Presentation Skills module in Intrepid and refer back to it often as you practice your presentation craft.

Some Helpful Links:

- ServiceNow CSDM vs. CMDB A City
Analogy: <https://www.lukaszszumilas.com/servicenow-csdm-vs-cmdb-a-city-analogy/>
- ServiceNow community video – CSDM in a nutshell: <https://www.youtube.com/watch?v=EnlhMhP2FsM>

(This is based on CSDM V3)

- John Spirko's CSDM presentation John Spirko
- <https://youtu.be/z8hG4eTRef4?si=ixNGk56lYjjiwT79>
- John Spirko's Digital Services Forum Community Page: click here
- Community video – CSDM Discussion: Technical Service vs. Business Services
<https://youtu.be/jWfwTHTLzl>

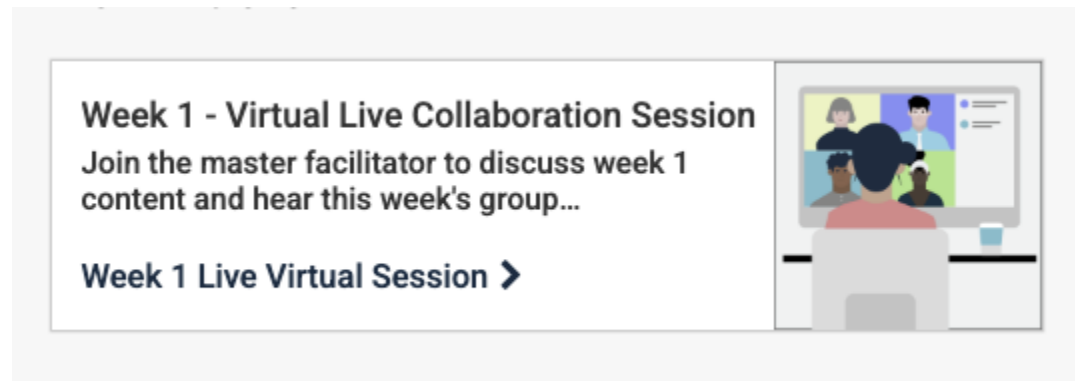
(based on CSDM V4)

- SDM to BIAN mapping (Others available. TOGAF, IT4IT)
- <https://www.youtube.com/watch?v=LJINzbfxiQo>
- CSDM to TBM mapping - <https://www.youtube.com/watch?v=l1zjjmh6xt8>
- There are other CSDM videos in the series. Search youtube in the ServiceNow Community channel for “CSDM 4”.
- CSDM Ebook: <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/ebook/ebk-csdm-101-primer.pdf>

Weekly Study Notes (Week 1)

Architect Responsibilities & Presentation Skills

Remember you can access the recordings for both VCS session for week 1 via Intrepid in the Week 1 tab.



Planning Your Presentation

- The facilitator is acting like your customer/stakeholder.
- Where was Suzi in the team presentations? Remember that you are ServiceWow Consulting Services (SCS) coming in to present to Sovereign Bank. Suzi Matsui is the lead CTA at SCS and you will either be presenting as Suzi or as somebody shadowing Suzi closely.
- The purpose of this week's meeting is to establish trust with the customer. To achieve this, you can bring attention to certain things:
 - The breadth of experience across the members of your team
 - Experience with similar customers, projects and industries
- In future weeks, you can spend less time on the introduction in your presentations (maybe very briefly on the speaker) so that you can allocate more time to the week's content.
- On the structure of presentations – Try and adopt an approach that establishes attention and outlines the challenges for the Bank. Start with a high-level overview of the key challenges, then walk through each challenge and solutions/recommendation in turn providing more detail. This approach

gives a clear map for the structure of the overall presentation, and also makes it clear which response/solution goes with each challenge.

Session Content

- Keep in mind the 3 pillars of People, Process and Technology when doing presentations with this context. In most cases the solutions you propose will involve to a greater or lesser extent these three pillars.
- Aligning your presentation to outcomes and challenges will make the solution you're presenting more impactful and easier to relate to from your customer's point of view.
- It was good to see so many challenges across the different groups, with some common elements and solutions coming up around OCM, Governance (Sticking to OOTB) etc. There will be plenty more discussion about challenges and how to address them in future weeks of the program!
- When creating your slide responses, always have in your mind (how does this related to Sovereign Bank). Consider using some examples from the case study to illustrate your point.

Some Presentation Skills Tips

- Use the Presentation Skills module in Intrepid for some leading practice and suggestions. It's there for you for all future weeks so refer back to it often as you practice.
- Slides:
 - The use of colour and contrasting colours are great combined with simple, clean slides as a starting point
 - Simple graphics work and don't detract from what you're presenting and provide a good contrast from just text based.
 - Also use transitions within the content of your slides to punctuate and pace the content that is being delivered.
 - Avoid using length textual content as this is difficult to presenter to deliver and for the audience to remember.

- Slides are a supporting tool, and not the actual presentation.
- Be careful of speech that doesn't match the level of formality of the meeting
- Ensure your camera is on and framed correctly. Have your camera at eye level or slightly above you. It's a good idea to step back slightly so that the camera can see you from the top of your head down to your chest. If you use hand gestures, there will then be room for them to be visible in the camera frame.
- Use voice tone, speed and volume to emphasise the important parts of your message. It's a good idea to pause slightly after you've said something particularly meaningful so that it stands apart from the rest of the speech around it.
- The video auto-analysis tool in Intrepid is great for analysing your rate of speech, filler words etc. Use this to help you improve rather than any exact measure of good/bad. A good way of handling pause words is to specifically plan spaces for you to pause between ideas or phrases. Eventually you can learn to replace your filler words with a short pause.
- If your space allows, you might try standing during your presentation. Some people find that this allows them to be more engaged with their audience since it's more similar to presenting in-person. In my opinion, standing can also raise your confidence levels.
- Don't like Presenter Mode in Powerpoint?
 - Try setting up a slide show (Slide Show → Set up Slide Show)
 - Select "Browsed by an individual (window)"
 - This displays the presentation in its own window that you can share on Zoom, and you can still have your outline/speaker notes in another window.

Some Helpful Links:

- <https://www.presentationzen.com/>: Site with good examples on how to improve your presentation and also a book around the subject.

- <https://www.Plexicam.com>: tool to have your webcam in the middle of the screen and get a more focus on your eyes when presenting. Not necessary but might consider it for the future.
- <https://www.youtube.com/watch?v=eRkgK4jfi6M>: This is a nice video to illustrate different levels of presentation depending on the audience.
- <https://www.youtube.com/watch?v=NEJzPeyqODo>: 7 ways to minimise resistance to change